



CLAUDE CERTIFIED ARCHITECT — FOUNDATIONS • DOMAIN 1

Building Agentic Applications with the Claude Agent SDK

A step-by-step technical walkthrough: the agent loop • tool & MCP integration • multi-agent orchestration • subagent delegation • lifecycle hooks • production controls

AAA Learning Group • engineering deep-dive • sources cited inline ([↗](#)) throughout

What we'll cover — and how to read this deck



Framing

Workflows vs agents; when NOT to build one

01



The agent loop

stop_reason, turns, message types, controls

02



Tools & MCP

custom tools, ACL design, scoping

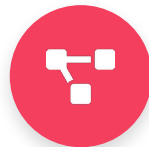
03



Orchestration

coordinator + subagent delegation

04



Lifecycle hooks

deterministic control & normalization

05



Production

budget, compaction, sessions, observability

06

These six blocks map directly to Domain 1 (Tasks 1.1–1.7) plus the Domain 2 / Domain 5 ideas the exam threads through agentic scenarios.

First principle: workflows vs agents

Anthropic's guidance: start with the simplest thing that works; add agency only when it demonstrably improves outcomes. Agents trade latency and cost for flexibility.



Workflow

LLM + tools orchestrated through predefined code paths.

- Predictable, consistent, debuggable
- Best for well-defined tasks
- You own the control flow



Agent

LLM dynamically directs its own process & tool use in a loop.

- Flexible, model-driven, scales to open-ended work
- Best when steps can't be predicted
- Higher cost; risk of compounding errors

Senior takeaway: “agentic” is a spectrum. Reach for full autonomy only for open-ended tasks in trusted environments with good feedback signals. Everything in this deck applies to both — the SDK runs the loop either way.

The composable patterns you'll combine



Prompt chaining

Fixed sequential steps; gate between them.

Use for: Predictable multi-aspect work



Routing

Classify input → specialized handler.

Use for: Distinct categories, separate prompts



Parallelization

Sectioning or voting, aggregated.

Use for: Speed or multiple perspectives



Orchestrator-workers

Coordinator splits & synthesizes dynamically.

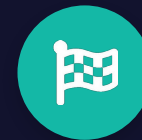
Use for: Subtasks can't be predicted



Evaluator-optimizer

Generate → critique → refine in a loop.

Use for: Clear criteria, iterative value



Where Domain 1 lives

The exam's multi-agent research system is orchestrator-workers; the support agent is a routed agent with enforcement. You'll build both today.

What the Claude Agent SDK gives you

The same runtime that powers Claude Code, as a library. It runs the loop and hands you control over tools, permissions, cost limits, sessions, subagents, and hooks. No CLI required.



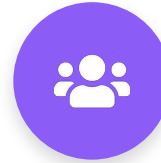
Runs the loop

Evaluate → call tools → feed results back → repeat.



Tools & MCP

Built-in tools + custom tools via in-process MCP.



Subagents

Delegate to isolated, specialized agents.

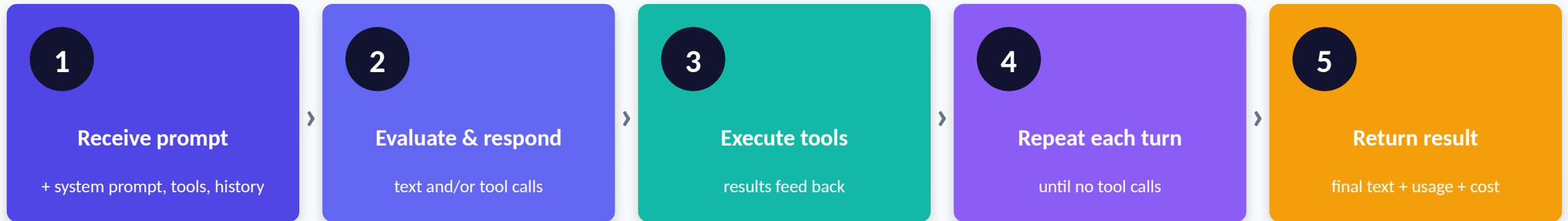


Control plane

Permissions, budget, effort, hooks, sessions.

Python `pip install claude-agent-sdk` **TypeScript** `@anthropic-ai/claude-agent-sdk` (renamed from “Claude Code SDK”, Sept 2025)

The agent loop, step by step



One turn = Claude produces output with tool calls → SDK executes them → results return automatically, without yielding to your code. Steps 2–3 loop until Claude responds with **no tool calls**.

Example — “Fix the failing tests in auth.ts”: Turn 1 Bash npm test → Turn 2 Read files → Turn 3 Edit + re-run (green) → final text-only turn. Four turns, three with tools.

Reading the loop: message types

As the loop runs the SDK yields a stream of typed messages. Knowing them is how you build progress UIs, audit trails, and cost tracking.

● **SystemMessage** Session lifecycle — subtype init / compact_boundary / ...

● **AssistantMessage** After each Claude response — text + tool-call blocks

● **UserMessage** After each tool execution — the tool result sent back

● **StreamEvent** Raw streaming deltas (only when partial messages on)

● **ResultMessage** End of loop — final text, token usage, cost, session_id

Drive most apps off ResultMessage (output + cost + subtype). Stream AssistantMessage for live progress.

Walkthrough 1 — the loop on stop_reason

```
messages = [{"role": "user", "content": prompt}]
for _ in range(max_safety_turns):    # circuit breaker
    r = client.messages.create(model, tools=TOOLS,
                               messages=messages)
    messages.append({"role": "assistant",
                    "content": r.content})
    if r.stop_reason == "end_turn":   # 2 DONE
        return final_text(r)
    if r.stop_reason == "tool_use":   # 1 CONTINUE
        results = run_tools(r.content)
        messages.append({"role": "user",
                        "content": results}) # 3
```

1

Branch on stop_reason

tool_use → run tools and loop again.

2

Terminate on end_turn

Return the model's final text. Not on text/iteration count.

3

Feed results back

Append tool_result blocks (keyed by tool_use_id) as a user turn.

Loop control: correct vs anti-patterns



Correct practice

- Branch on stop_reason every turn
- Append tool results to history so the model reasons over them
- Let Claude choose the next tool from context (model-driven)
- Keep a high turn/budget cap purely as a circuit breaker



Anti-patterns

- Parse natural-language text to decide it's "finished"
- Use an iteration cap as the PRIMARY stop mechanism
- Treat any assistant text as a completion signal
- Hard-code a fixed tool sequence / decision tree

Controlling the loop in production



max_turns

Cap tool-use round trips. Prevents runaway sessions on open-ended prompts.



max_budget_usd

Stop at a spend threshold. The recommended default for production agents.



effort

low → max reasoning per turn. Use low for file lookups; high for refactors/debugging.



permission_mode

default / acceptEdits / plan / dontAsk / bypassPermissions — gate what runs without asking.

Hit a limit and the loop ends with a ResultMessage subtype (error_max_turns / error_max_budget_usd) — capture session_id and resume with a higher cap.

Handling the result — every termination state

● success

Task finished normally — result text present

● error_max_turns

Hit maxTurns — resume with a higher cap

● error_max_budget_usd

Hit budget — decide whether to extend

● error_during_execution

API failure / cancellation interrupted the loop

● error_max_structured_output_retries

No valid structured output within retry limit

Always check subtype before reading result. All subtypes carry `total_cost_usd`, `usage`, `num_turns`, `session_id` — so you can track spend and resume even after a failure.

Tool integration — step by step



1

@tool

Name, description, input schema,
async handler.



2

create_sdk_mcp_server

Bundle tools into one in-process
MCP server.



3

allowed_tools

Pre-approve
mcp__<server>__<tool> names.



4

is_error

Return failures as data the loop can
recover from.

Tools run in-process inside your app — your database client, your API auth, your domain logic. Scope each agent to ~4–5 tools; selection reliability degrades as the set grows.

Walkthrough 2 — a custom tool

```
@tool(
    "lookup_order",
    "Look up an ORDER by order ID (0-###).",
    "Returns total, status, customer. NOT for"
    "customer lookups.", # 1 differentiating desc
    {"order_id": str}) # 2 input schema
async def lookup_order(args): # 3 handler
    rec = _ORDERS.get(args["order_id"])
    if not rec:
        return {"content": [...],
                "is_error": True} # 4 recoverable
    return {"content": [{"type": "text", ...}]}
```



Description = selection signal

Distinct, bounded text. The model routes on this.



Namespaced + scoped

mcp__support__lookup_order in allowed_tools.



Structured errors

is_error keeps the loop alive so the agent recovers.

Agent-computer interface (ACI) design

Invest in your tool interface as much as your prompt. “Put yourself in the model's shoes” — a tool should be as obvious to Claude as a great docstring is to a new teammate.



Rich descriptions

Input formats, example queries, edge cases, and explicit boundaries vs similar tools — descriptions are the primary selection mechanism.



Split, don't overload

Break a generic `analyze_document` into `extract_data_points`, `summarize_content`, `verify_claim`. Rename to remove overlap.



Scope per agent

~4–5 tools per role. Tools outside an agent's specialization get misused (a synthesis agent that web-searches).



Poka-yoke the args

Make mistakes impossible: require absolute paths, validate URLs, constrain enums. Anthropic spent more time on tools than prompts for SWE-bench.

Built-in tools, MCP, and context cost



File & search

Read · Write · Edit · Glob · Grep



Execution & web

Bash · WebSearch · WebFetch



Orchestration

Agent · Skill · AskUserQuestion · TaskCreate/Update



External / custom

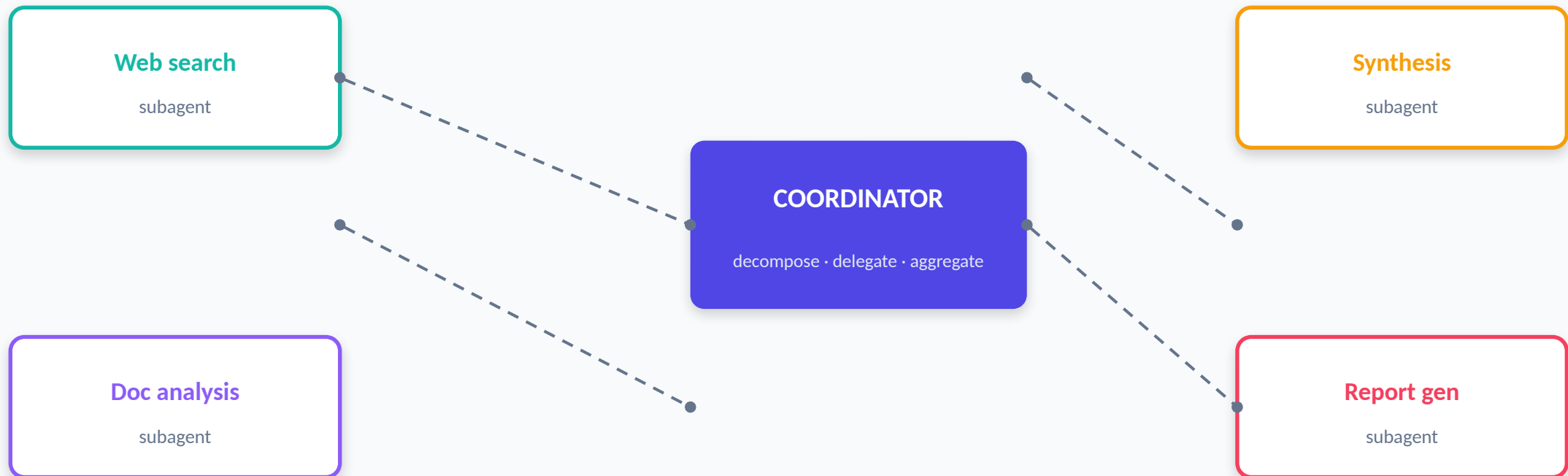
MCP servers + your @tool handlers

Context is a budget you spend before the agent does any work

- Every tool definition consumes context on every request — another reason to scope tightly.
- MCP tool search defers MCP schemas and loads them on demand; without it, each server adds all its schemas up front.
- Read-only tools (Read/Glob/Grep, MCP tools marked read-only) can run in parallel; state-changing tools run sequentially.
- Mark custom tools `readOnlyHint` to allow parallel execution.

Multi-agent orchestration: hub & spoke

A coordinator owns ALL inter-agent communication, routing, and error handling. Subagents never talk to each other — everything flows through the hub for observability and control.



What a good coordinator does



Dynamic selection

Analyze the query and invoke only the subagents it needs — don't always run the full pipeline.



Broad decomposition

Partition the WHOLE topic into non-overlapping subtasks. Narrow decomposition silently drops coverage.



Iterative refinement

Evaluate synthesis for gaps, re-delegate targeted queries, re-synthesize until coverage is sufficient.



Route everything

All subagent comms go through the hub for observability and consistent error handling.

Walkthrough 3 — subagent delegation

```
options = ClaudeAgentOptions(  
    allowed_tools=["WebSearch", "Task", "Agent"], # 1  
    agents={  
        "web-researcher": AgentDefinition( # 2  
            description="Searches ONE subtopic; cited.",  
            prompt="Return claim+excerpt+URL+date...",  
            tools=["WebSearch", "Read"], # 3 scoped  
            model="sonnet", effort="medium"),  
        "synthesizer": AgentDefinition(  
            description="Combines findings into report.",  
            tools=["Read"]), # no web -> can't go rogue  
    })
```

1

Include the spawn tool

Coordinator allowed_tools MUST list Task/Agent.

2

AgentDefinition

description, prompt, tools, model, effort, skills, mcpServers.

3

Scope per role

Researcher searches; synthesizer can only Read.

Subagents run with isolated context

A subagent starts a FRESH conversation. The only channel from parent to child is the spawn prompt string — pass every file path, finding, and decision it needs directly in that prompt. Only the subagent's final message returns to the parent.



Subagent RECEIVES

- Its own system prompt (AgentDefinition.prompt)
- The spawn prompt string from the parent
- Tool definitions (all, or the scoped subset)
- Project CLAUDE.md (if setting sources enabled)



Subagent does NOT receive

- The parent's conversation history or tool results
- The parent's system prompt
- Skills (unless listed in the definition)
- Any shared live memory between invocations

Context isolation is the FEATURE: a subagent can read 30 files and only a concise summary returns — the parent's context stays lean. Subagents also cannot spawn their own subagents.

Spawning: the Task→Agent rename + parallelism



Know both names

The exam guide calls the spawn tool **Task** and requires allowedTools to include "**Task**". It was renamed to **Agent** in Claude Code v2.1.63; current SDKs emit "Agent" in tool_use blocks but still list "Task" in system:init. **Include and match BOTH.**

Spawn in parallel — emit multiple spawn calls in ONE coordinator response

Parallel ✓

One turn → 3 spawn calls → run concurrently. Big latency win for independent searches.

```
[Task(a), Task(b), Task(c)] # single response
```

Sequential ✗

One spawn per turn forces round-trips and slows the whole pipeline unnecessarily.

```
turn1: Task(a); turn2: Task(b); ...
```

Passing context & preserving provenance



Pass complete findings

Include prior agents' outputs directly in the next subagent's prompt (search + analysis → synthesis).



Separate content from metadata

Carry source URL, doc name, page, and date alongside each claim so attribution survives synthesis.



Goals over procedures

Coordinator prompts state research goals & quality criteria, not rigid scripts — so subagents adapt.



Structured handoffs

On escalation, compile a summary (IDs, root cause, recommended action) for humans without the transcript.

Failure mode: narrow decomposition

Scenario: Topic “impact of AI on creative industries.” Every subagent succeeds, yet reports cover only visual arts. Logs show the coordinator split it into *digital art, graphic design, photography* — music, writing, and film were never assigned.



Root cause: the coordinator's task decomposition was too narrow. The subagents executed correctly within the scope they were given.

Why not the distractors

- Synthesis agent — it faithfully synthesized what it was given
- Web-search agent — it found relevant articles for its assigned slices
- Doc-analysis agent — it summarized correctly within scope

Diagnostic heuristic: when every worker succeeds but coverage is wrong, suspect the decomposition — not the workers.

Lifecycle hooks — control points

Hooks are callbacks that fire at agent events. They run in YOUR process (not the model's context) and give deterministic control where prompts can only ask nicely.

EVENT	FIRES	USE FOR
 PreToolUse	Before a tool runs	Allow / deny / modify input — block refunds > \$500
 PostToolUse	After a tool returns	Normalize data, append context, replace output
 SubagentStart / Stop	Subagent lifecycle	Track parallel spawning; aggregate results
 UserPromptSubmit	On user prompt	Inject extra context into the prompt
 Stop / PreCompact	End / before compaction	Save state; archive transcript before summarizing

Walkthrough 4 — PreToolUse enforcement

```
async def enforce_refund_limit(input, id, ctx):
    if input["tool_name"]=="mcp__ops__process_refund":
        amt = float(input["tool_input"]["amount"])
        if amt > 500:                                # 1 rule
            return {"hookSpecificOutput": {
                "hookEventName": input["hook_event_name"],
                "permissionDecision": "deny",           # 2 block
                "permissionDecisionReason":             # 3
                    "Refunds > $500 need a human.{}".format(id)}
            }
        return {} # allow everything else
```

1

deny blocks the call

permissionDecision: allow | deny | ask | defer.

2

Always give a reason

The model sees it and won't blindly retry.

3

deny wins

If any hook denies, the call is blocked. Hooks run in parallel.

PreToolUse can also rewrite a call: return updatedInput (with permissionDecision "allow") to sandbox a path or sanitize arguments before execution.

PostToolUse — normalize heterogeneous data

Different MCP tools return dates as Unix timestamps, ISO-8601, or numeric codes. A PostToolUse hook normalizes results BEFORE the model reasons over them — removing a whole class of confusion.

```
async def normalize_timestamps(input, id, ctx):
    if input["tool_name"]=="mcp__ops__lookup_order":
        data = json.loads(raw_text(input))
        iso = to_iso8601(data["ordered_at"]) # 1719500000→
        return {"hookSpecificOutput": {
            "hookEventName": input["hook_event_name"],
            "additionalContext": # append
                f"Normalized ordered_at (ISO): {iso}"}}
    return {}
```

Fires after execution

Transform results the model hasn't seen yet.

additionalContext

Append normalized info to the tool result.

updatedToolOutput

Replace the raw output entirely when needed.

Hooks vs prompts: deterministic vs probabilistic



Hooks / prerequisite gates

- Deterministic — the rule ALWAYS holds
- Code-enforced, outside the model
- Right for must-hold business rules
- e.g. block refunds > \$500; verify identity first



Prompts / few-shot

- Probabilistic — non-zero failure rate
- Great for judgment, tone, and guidance
- NOT sufficient when errors cost money
- Use to shape behavior, not to guarantee it



Exam rule of thumb: if a business rule MUST hold (identity before refund, dollar limits, ordering of steps), enforce it with a hook or prerequisite gate — never with a stronger prompt or more examples. This single distinction resolves a large share of Domain 1 scenario questions.

Walkthrough 5 — programmatic prerequisite gate

Problem: in 12% of cases the agent skips `get_customer` and looks up orders by stated name → wrong account, wrong refund. The fix is a programmatic prerequisite, not a stronger prompt.

```
VERIFIED = set()    # state OUTSIDE the model
GATED = {"mcp__support__lookup_order",
         "mcp__support__process_refund"}

async def require_verification(input, id, ctx):
    if input["tool_name"] in GATED and not VERIFIED:
        return deny("Call get_customer to verify first")
    return {}
```



A ✓ Prerequisite gate

Deterministic ordering — correct.



B ✗ “Mandatory” prompt

Probabilistic; still fails sometimes.



C ✗ Few-shot examples

Probabilistic; doesn't guarantee.



D ✗ Routing classifier

Changes availability, not ordering.

Context window & automatic compaction

Context does not reset between turns — system prompt, tool defs, history, and tool outputs all accumulate. Large tool outputs are the usual culprit.



Auto-compaction

Near the limit, the SDK summarizes older history (emits a `compact_boundary` message). Early instructions may be lost.



Persistent rules → CLAUDE.md

Re-injected every request, so they survive compaction — unlike instructions buried in the first prompt.



Subagents for heavy work

Isolate verbose exploration; only the summary returns to the parent's context.



Trim & tune

Keep only relevant tool fields; lower effort for routine turns; PreCompact hook to archive transcripts.

Sessions: continue, resume, fork



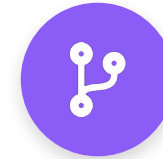
Continue

Pick up the MOST RECENT session in the directory. No ID tracking. One conversation at a time.



Resume

Return to a SPECIFIC session by ID. Needed for multi-user apps or recovering after a limit/restart.



Fork

Branch a COPY of history to try a different approach; the original stays untouched. Two independent IDs.

Production notes

- Capture `session_id` from `ResultMessage` (present even on errors). Python's `ClaudeSDKClient` tracks it for you.
- Sessions persist the CONVERSATION, not the filesystem — use file checkpointing to revert file changes.
- Resuming across hosts: move the `.jsonl` (cwd must match), or — often more robust — inject a structured summary into a fresh session.

Error propagation across agents

When a subagent fails (e.g. web search times out), return STRUCTURED error context so the coordinator can recover intelligently — retry, try an alternative, or proceed with partial results.



Return structured context ✓

- failure type (timeout, validation, permission)
- the attempted query
- any partial results gathered
- possible alternative approaches
- isRetryable flag for the coordinator



Anti-patterns ✗

- Generic “search unavailable” — hides context
- Empty results marked as success — silent failure
- Kill the entire workflow on one failure
- Confuse access failures with valid empty results

Operating agents in production



Observe

Stream message types; SubagentStart/Stop + PostToolUse hooks for audit trails. Hooks don't cost model context.



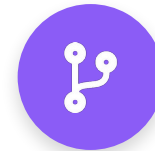
Bound cost

max_budget_usd + per-turn effort; read total_cost_usd / usage off every ResultMessage.



Constrain

permission_mode + allow/deny rules + PreToolUse gates. bypassPermissions only in isolated envs.



Recover

Capture session_id; resume on limits; structured errors; checkpoints for file rollback.

Reference architectures



Customer support agent

Single routed agent · MCP tools · 80%+ first-contact resolution

- Tools: get_customer, lookup_order, process_refund, escalate_to_human
- Prerequisite gate: verify identity before order/refund ops
- PreToolUse hook: cap refunds; redirect to escalation
- Escalate on explicit request, policy gaps, no progress
- Persist a 'case facts' block; trim verbose tool output



Multi-agent research

Orchestrator-workers · search / analysis / synthesis / report

- Coordinator decomposes broadly; spawns searchers in parallel
- Subagents: isolated context, scoped tools, cited findings
- Pass findings + metadata explicitly to synthesis
- Structured error context → recover with partial results
- Annotate coverage gaps and conflicting sources

Exam cheat sheet — traps & truths

Loop control

Drive on stop_reason — not text, not iteration caps.

Spawning

Coordinator needs Task/Agent; subagents can't recurse.

Parallelism

Multiple spawn calls in ONE response.

Tool count

Scope tightly; ~4–5 beats 18 for reliability.

Production

Set max_budget_usd; check ResultMessage subtype.

Tool results

Append as a user turn with tool_result + tool_use_id.

Subagent context

Isolated — pass everything in the spawn prompt.

Hooks vs prompts

Must-hold rule → hook; judgment → prompt/few-shot.

Errors

Structured context > generic 'failed'; is_error to recover.

Continuity

Stale results → fresh session + summary, not resume.

Hands-on lab — code walkthrough files



01_agentic_loop_raw_api.py

Bare-metal loop on stop_reason.



02_custom_tools_sdk.py

@tool + in-process MCP server.



03_coordinator_subagents.py

Coordinator + scoped subagents.



04_lifecycle_hooks.py

PreToolUse block + PostToolUse normalize.



05_prerequisite_gate.py

Deterministic verify-before-refund gate.



Agent_SDK_Walkthrough.ipynb

Colab notebook — run all five live.

Try it: `pip install anthropic claude-agent-sdk` · set `ANTHROPIC_API_KEY` · then run each file or open the notebook in Colab. Pair with the 15-question MCQ set.

KEY TAKEAWAYS

Six things to carry into the exam — and into prod

- Choose the simplest architecture; agency is a cost/latency trade, not a default.
- The agent IS the loop — drive on `stop_reason`; the SDK runs it, you control it.
- Orchestrate hub-and-spoke; coordinators decompose broadly and route everything.
- Subagents = isolated context + scoped tools; pass what they need explicitly.
- Hooks give deterministic guarantees; use them whenever a rule must always hold.
- Operate it: budget, permissions, observability, sessions, structured errors.



Go deeper: platform.claude.com/docs/en/agent-sdk (overview · agent-loop · subagents · hooks · custom-tools · sessions) · anthropic.com/engineering (building effective agents · multi-agent research). Full annotated list in `Reference_Links.md`.



Questions & discussion

Next: run the Colab lab → take the 15-question MCQ set → bring edge cases back to the group.